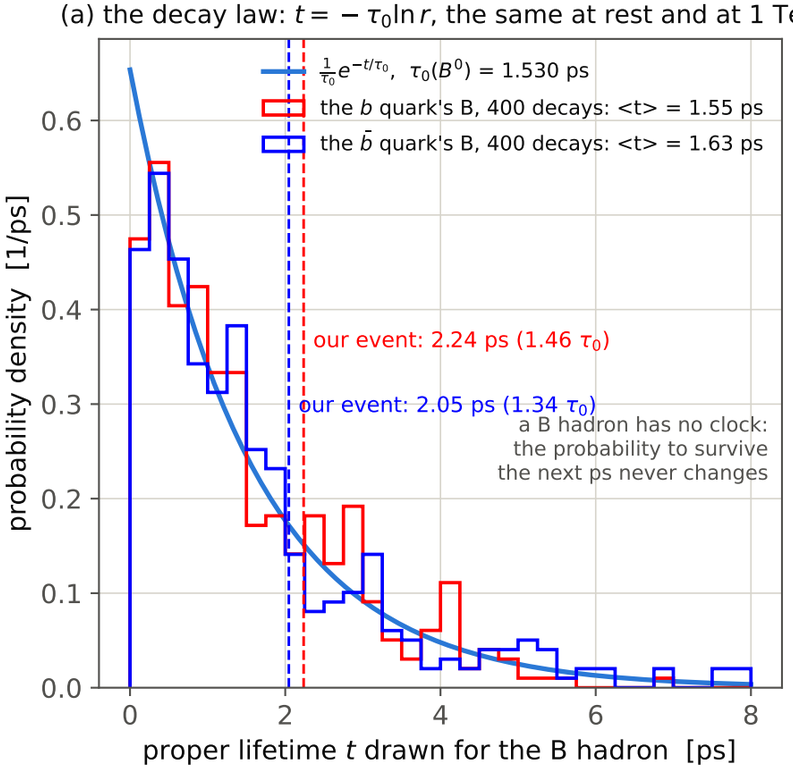
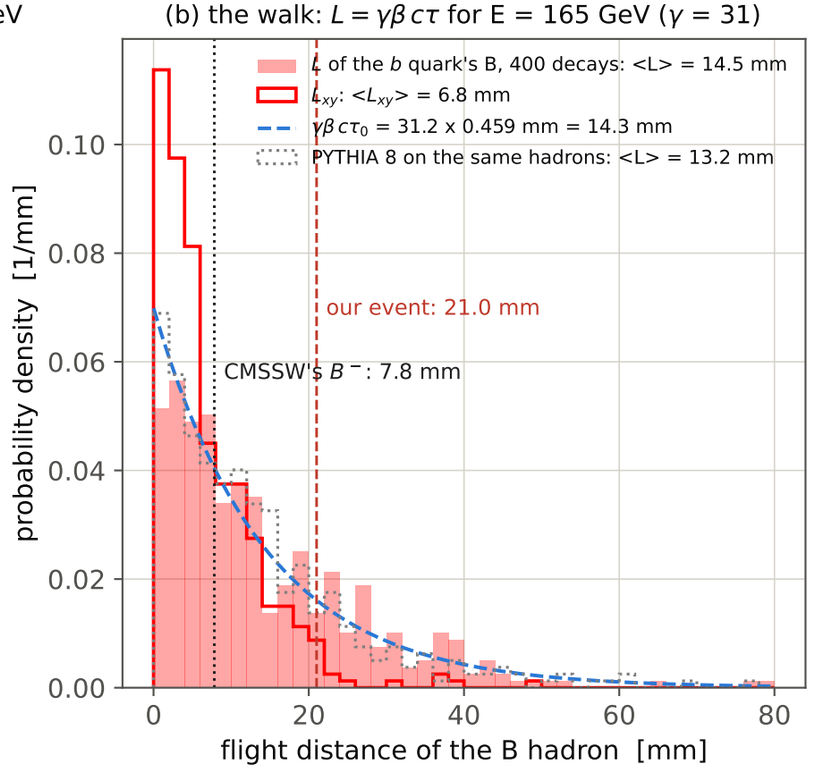
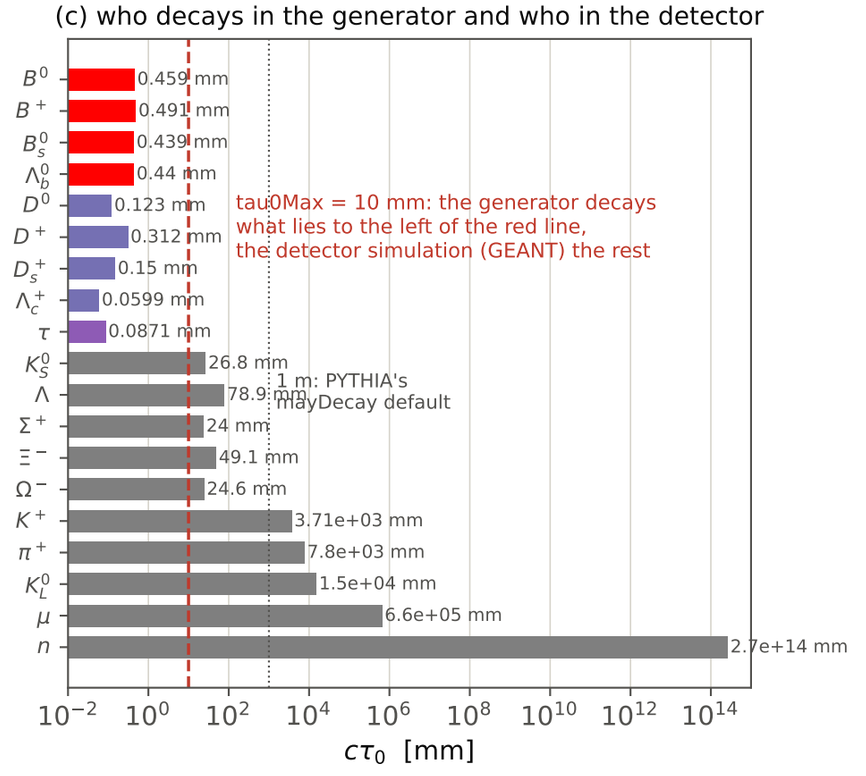
Lifetimes: an exponential clock, a walk of a centimetre at the LHC, and the 10 mm rule of the generator (page 1)

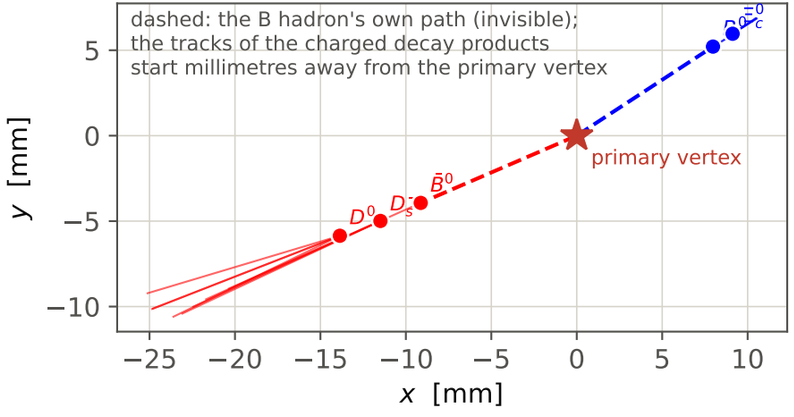




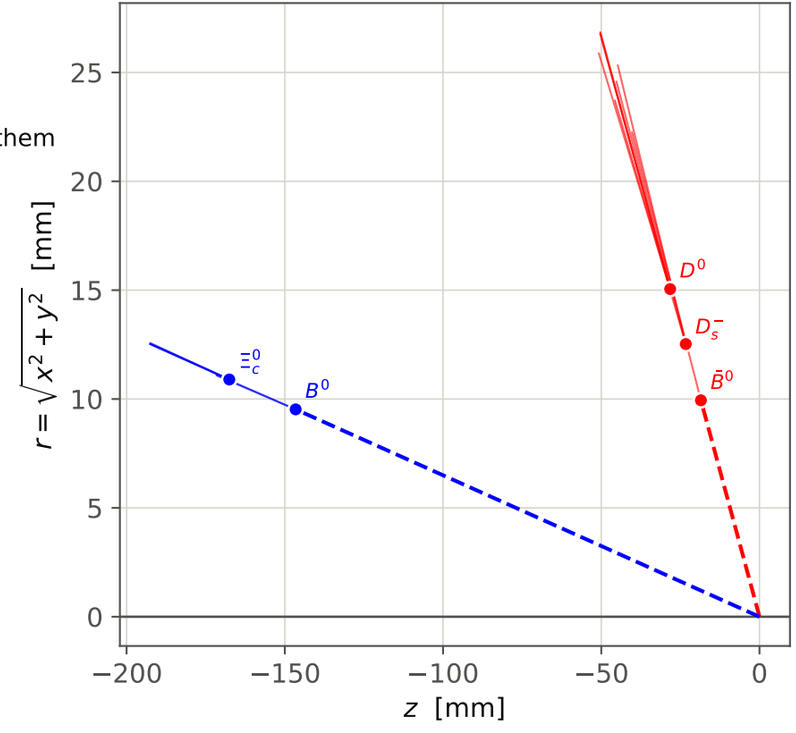


Displaced vertices: the B walks, its daughters' tracks miss the primary vertex by hundreds of microns (page 2)

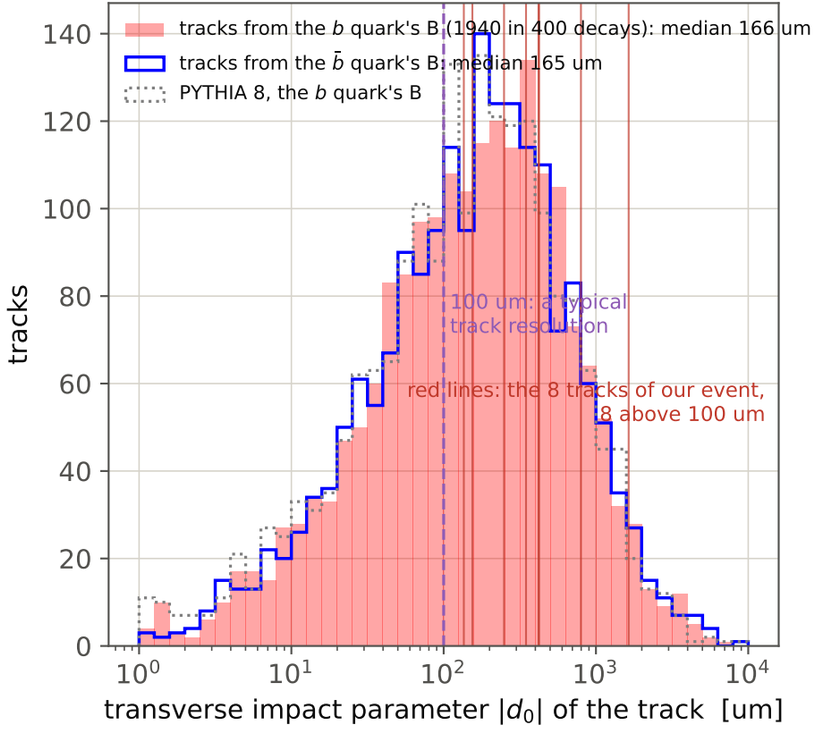
(a) decay vertices of our event in (x, y) and the tracks leaving them



(b) the same along the beam: the $\bar{b}$ hadron (η = -3.4) walks far in z

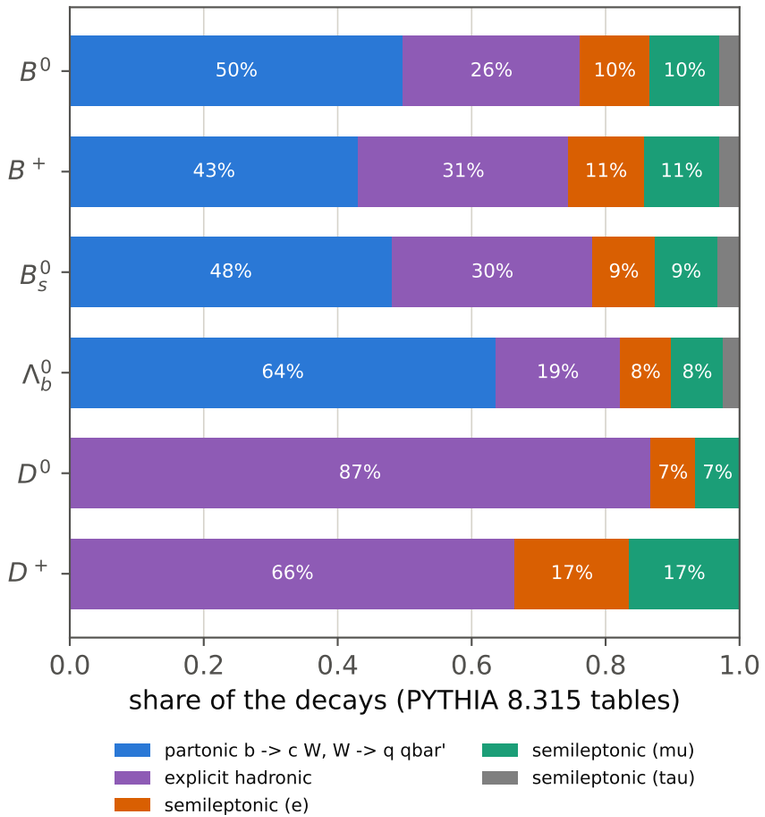


(c) the birthmark: impact parameters of the tracks from the B



The decay tables: 870 channels for a B^0, half of them written at quark level (page 3)

(a) what the tables say



(b) the decay chains of our event (branching ratio of each step)

the b quark's hadron ($\bar{B}^{*0}$ row 1012, p_T = 78.6 GeV)

$\bar{B}^{*0} \to \bar{B}^0\ gamma$ (100.0%)
- $\bar{B}^0 \to D_s^{*-}\ D^{*0}\ \rho^+$ (7.4%) L = 21.01 mm
  - $D_s^{*-} \to D_s^-\ gamma$ (94.2%)
    - $D_s^- \to K^{*-}\ K^{*0}$ (7.0%) L = 5.44 mm
      - $K^{*-} \to \bar{K}^0\ \pi^-$ (66.6%)
      - $K^{*0} \to K^+\ \pi^-$ (66.5%)
  - $D^{*0} \to D^0\ gamma$ (38.1%)
    - $D^0 \to K^-\ \omega\ \pi^+$ (3.0%) L = 10.98 mm

15 final particles (8 charged, 6 photons, 0 neutrinos)

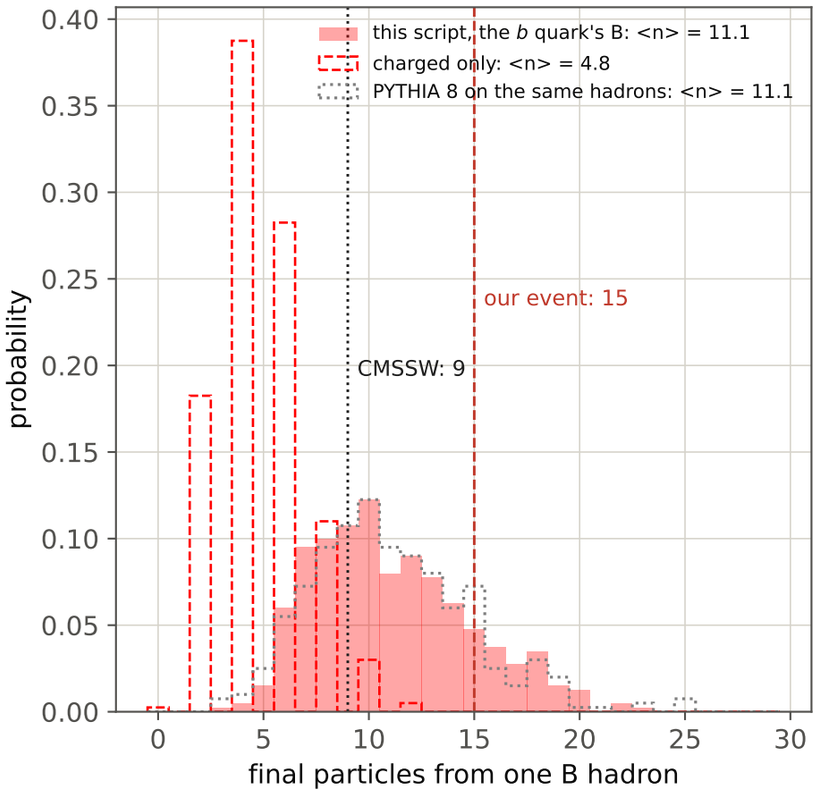
the $\bar{b}$ quark's hadron (B^{*0} row 1069, p_T = 82.8 GeV)

$B^{*0} \to \bar{B}^0\ gamma$ (100.0%)
- $\bar{B}^0 \to \Xi_c^0\ \pi^+\ \bar{\Delta}^-$ (0.6%) L = 146.88 mm
  - $\Xi_c^0 \to \bar{K}^{*0}\ \pi^-\ \Sigma^{*+}$ (10.0%) L = 21.02 mm
    - $\bar{K}^{*0} \to K^-\ \pi^+$ (66.5%)

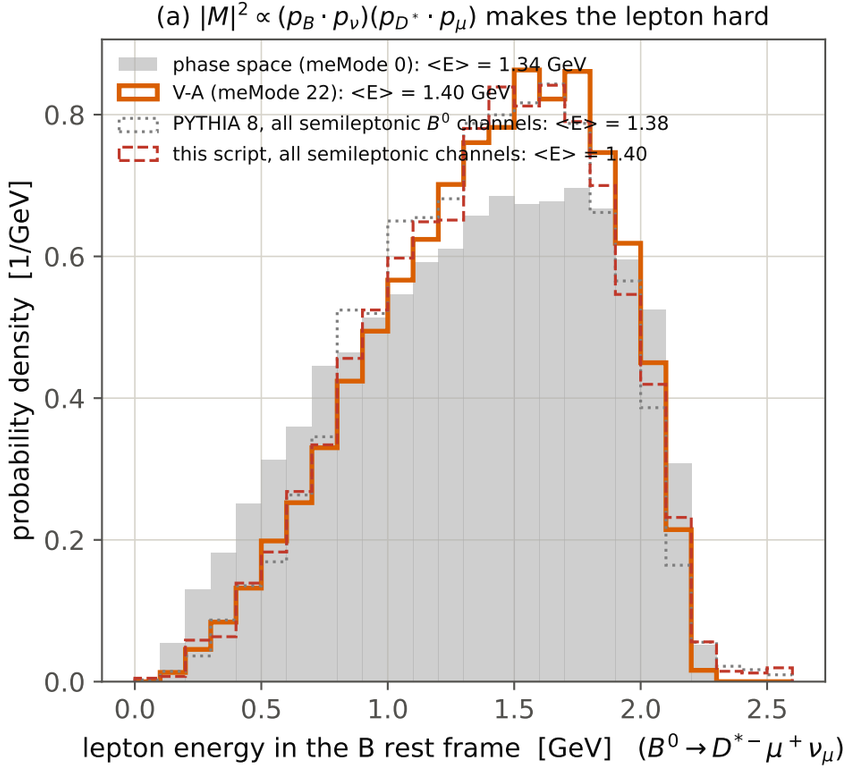
9 final particles (6 charged, 2 photons, 0 neutrinos)

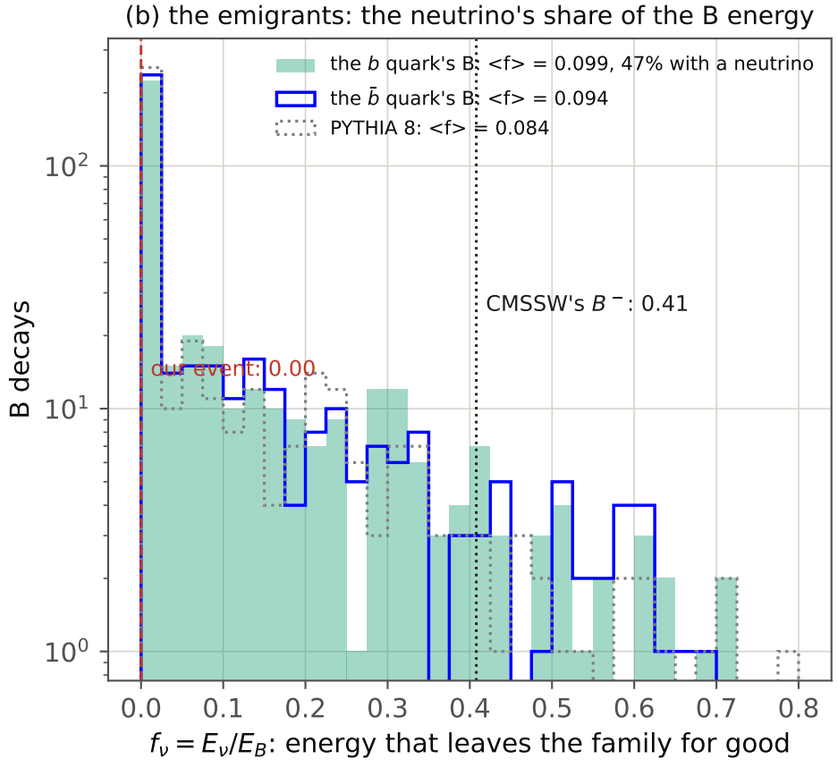
CMSSW's event: $B^- \to \bar{\nu}_e e^- D_0^{*0}$ (9 final, a 37.5 GeV neutrino);
$B^{*+} \to B^+ \gamma$, $B^+ \to \bar{D}^{*0} \pi^+ \pi^0 K^0 \bar{K}^{*0}$ (16 final)

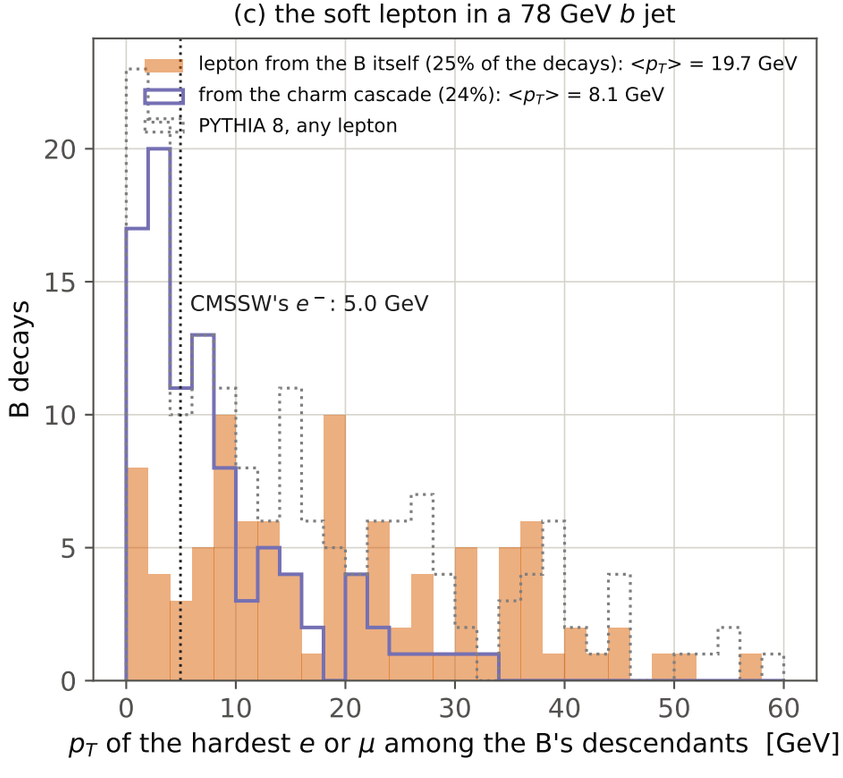
(c) how many grandchildren a B hadron leaves



Semileptonic decays: a hard lepton, a harder neutrino, and the energy the jet will never recover (page 4)

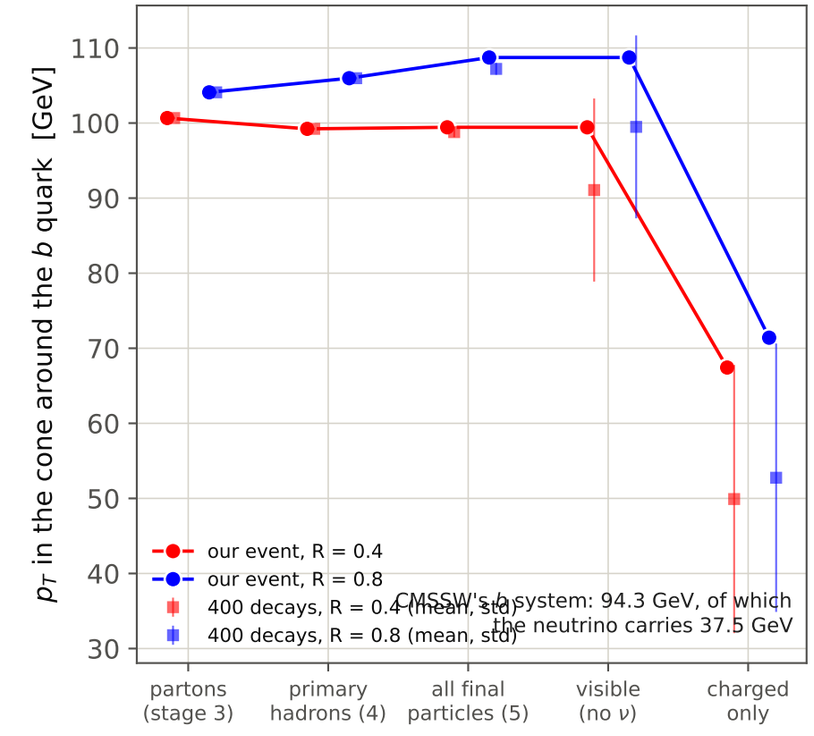




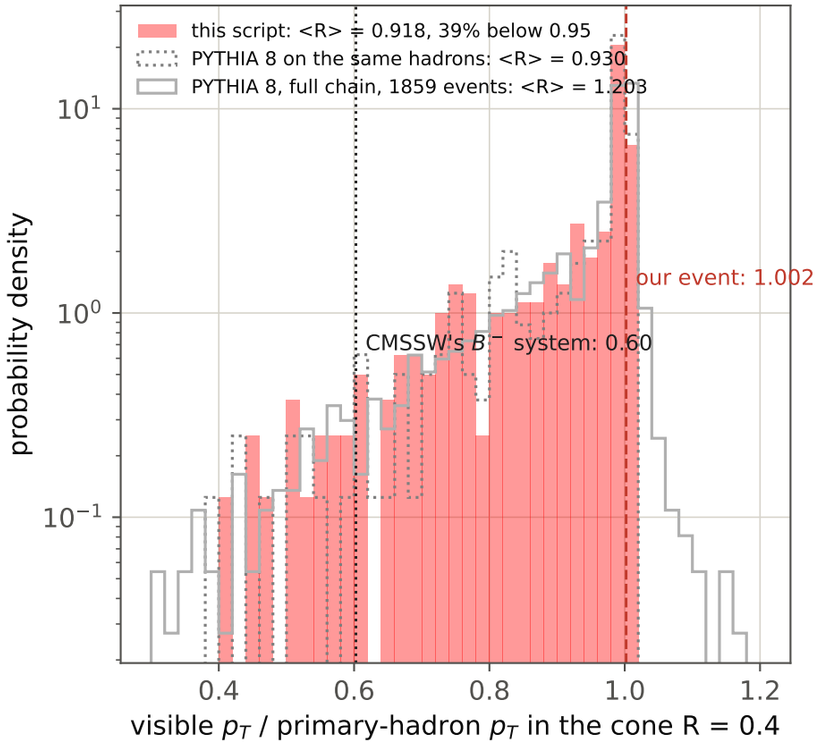


Our jet at this stage: 100.67 -> 99.23 -> 99.43 -> 99.43 GeV within R = 0.4 (page 5)

(a) the cone through the stages: decays keep the sum, neutrinos take theirs



(b) the response of the jet after the decays



(c) four times more particles, the same mass

